1 Data Structure of Operations

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This document describes each of the fields for an operation. In this case, the index tabs specify how the fields are structured. The actual sequence may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

General tab

Order / work plan

The order number or rather the work plan number is an upper-level number, under which each of the operations is compiled.

Sequence

The sequence number is the number of operation sequences in use.

OP

The operation number is the number listed below the order used to identify the operation.

Split

The split number.

OP name

Name of the operation; generally simply a short description of the activities that will be performed.

Article/Item

Part/item number of the article or material that is produced with the operation. If you do not enter the article, the system takes the value from the corresponding field of the order header.

Drawing issue number

Drawing issue number of the article, also referred to as index (available as of BDE 8.2).

Material type

Material type of the article that is to be produced in this particular production step. If you do not enter the material type, the system takes the value from the corresponding field of the order header.

Priority

You can use the "priority" as a control tool. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

Depending on the Order type, you can configure the priority to refer either to the operation or to the order. Choosing the latter will mean that the system will take the priority of the operation from the order header.

Planned on

This identifier indicates whether the operation is located in the group pool or if it is planned on a specific workplace / machine . In MES, you can plan operations either in the graphic detailed planning or via the order sequencing.

Entering or deleting the workplace later will NOT automatically change this identifier.

Planned workplace

If you set the identifier **planned on workplace**, this means that the operation is planned for the workplace entered here. If the input field is empty, the operation is not planned for any workplace.

<u>Please note:</u>

When you log on an operation, this field automatically includes the workplace where you logged in the operation. Doing so will overwrite any (in some cases a different) workplace for which the operation was planned up until then. As a result, the OP is implicitly re-planned.

Group

(Planned) station / machine group designated for producing the operation. It is meant as a planning criterion for group-oriented planning and in the graphic detailed planning.

If you log on an operation to a (different) workplace, its group will be updated, if necessary.

If, due to logging in the operation, there is a change to the group for which the operation was planned up until now, NONE of the values is taken from the template (this only happens if modifications are made manually via the editing function).

Fixed

This identifier specifies whether an operation is set as fixed during the planning process.

Before running automatic planning, the capacities (workplaces) are completely released with the exception of the fixed operations. Fixed operations that are still set in the past are moved to the right and set to "now" at the earliest plus a planning lead time. Any (fixed) operations planned for the future remain dispatched without changes.

Material

This field includes the first resource of the type "Material" (ID: "MAT") available in the component list. This is the "most important" input material.

You can use this field in planning, for example, when applying the <u>Setup change list</u> or in graphic detailed planning when planning equipment setup changes. It is of no significance to processing as part of the material and production logistics (MPL).

Color

You can enter the color of the main input material or the article planned for production here. This field is used in planning for example in the Setup change list or when planning equipment setup changes.

Tool

If you use the Tool and Resource Management (WRM) product group, this field includes the first resource available in the production resource / component list that is **not** of the resource type "DNC" or "MAT".

To this end, when a component is being entered that does **not** have a resource type "DNC" or "MAT", the system checks whether this field already includes a value. If the field does not include a value, this component is entered. For this reason, we recommend to first input the "main production resource" in the production resource list.

The graphic detailed planning integrates this field in order to identify production methods. However, the production resources stored in the operation are relevant when checking capacities.

By default, the field is of no relevance for processing in the Tool and Resource Management (WRM) product group.

DNC

If you use the production facility/resource management, this field includes the first resource available in the production resource / component list that is of the resource type "DNC" (ID: "DNC"). To this end, when a component with the resource type "DNC" is entered, the system checks whether this field already includes a value. If the field does not include a value, this component is entered.

In the system, this field is mainly used as a comment. By default, the field has no significance for DNC processing.

Upload number

The purpose of the confirmation/upload number is to identify an operation. This is a numeric value used for postings as an alternative to the combined order/OP number.

Examples

- Most of the time, a bar code is hard to read if the order / OP number is long (for example when using handheld barcode readers with a limited scanning range);
- If the space available on the work document is not large enough.

Please note: The length of the input field depends on the settings made for "Length of upload/confirmation number" in the basic parameter settings. If you did not specify a length there, the field is shown across the whole width of the application.

Leave this field blank for work plan operations.

Authorization

An authorization identifier that indicates whether a user is authorized to log on/off the operation. This involves crosschecking the identifier *OP postings* in the HR master.

Cost type

The cost type to be posted when executing this operation, for example in an overhead cost operation / order. At the moment, this field is only used as a comment.

Cost center

The cost center to be debited when executing this operation, for example in an overhead cost operation / order. At the moment, this field is only used as a comment.

Dates tab

The following dates are results calculated and executed during lead time scheduling. Lead time scheduling is triggered by certain events and runs asynchronously in the MES.

Scheduled start time

Scheduled start date of the operation as a result of the lead time scheduling compared to infinite capacities.

As a rule, a fixed operation is never changed. If, based on the scheduling situation, a date cannot be maintained, the operation is rescheduled, but it remains fixed.

Scheduled end time

Scheduled end date of the operation as a result of the lead time scheduling compared to infinite capacities.

Earliest start

Earliest start date (EST) of an operation as a result of forward scheduling during lead time scheduling as compared to infinite capacities or specified by PPS.

Earliest end

Earliest end date (EET) of an operation as a result of forward scheduling during lead time scheduling as compared to infinite capacities or specified by PPS.

Latest start

Latest start date (LST) of the operation as a result of backward scheduling during lead time scheduling as compared to infinite capacities.

Latest end

Latest end date (LET) of the operation as a result of backward scheduling during lead time scheduling as compared to infinite capacities.

Buffer time

The system determines the buffer time from the difference between the latest start date (LST) and the earliest start date (EST) for an operation

The sum total of the buffer times of all operations is stored in the order (header) in the field *OP buffer.*

Reducible time

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available (basic end date exceeded), then the MES will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the wait times and the transport times.

This value indicates how many (more) hours can be reduced from the lead time of an order. This time results from the sum of the:
- difference from the current waiting time and the minimum waiting and the
- difference from the current transport time and the minimum transport time
Reducible time = (current waiting time - minimum waiting time) + (current transport time - minimum transport time). These differences are displayed here as totals.

The document entitled Reduction Strategies provides information on how to configure reduction strategies.

Planned start

Planned start date for the operation.

Logging in an operation that has not yet been planned will not result in the following:
- the time when this operation is started will not be interpreted as the planned start date and
- this date will not be entered here.

Planned end

Planned end date for the operation.

The planned dates (planned start/planned end) are set:

- by HYDRA Shop Floor Scheduling (HLS) upon saving
- by the Graphic Order Sequencing (GAV) upon saving
- by manual data maintenance (client)
- by the interface

The following logic applies for inserting and/or editing an operation (manual data maintenance or by interface):

- Insert/copy operation
    - If the "planned start" field is empty, it will be assigned to the earliest start date by default.
    - If the "planned end" field is empty, it will be assigned to the latest end date by default.

    - In both cases, however, the operation is not planned (automatically), i.e. you can still replan the operation.
- Change operation
    - In this case, processing depends on the "planning function" option of the workplace:
        - N (no planning):
            - If the "planned start" field is empty, it will be assigned to the earliest start date by default.
            - If the "planned end" field is empty, it will be assigned to the latest end date by default.
        - In any other cases, the planned dates will not be set automatically through processing.

Quantities tab

Generally, you can enter quantities in four different quantity units for an operation. Enter the target quantity and the unit (as abbreviation) for each quantity unit. You can also enter a calculated "estimated scrap" quantity. These quantities can be specified
- by the PPS system or, if the target quantity update is activated,
- they may result from the quantity produced by the previous operation.

The letters in parentheses behind the field descriptions provide information about the particular quantity type.

(P) Primary quantity unit

(S) Secondary quantity unit

(T) Tertiary quantity unit

(B) Base quantity unit

Target quantity (P) / unit / target scrap quantity (P)

Use the primary quantity to enter data via the terminal (primary input quantity).

The indicated target quantity may include a target scrap quantity that might have been entered.

Send-ahead quantity

In order to illustrate any overlapping, you can define a minimum send-ahead quantity (in the primary quantity unit) for the (preceding) operation. You can start the following operation (overlapping) if at least the send-ahead quantity has been finished and posted. The system verifies the send-ahead quantity during data collection (when logging on operations). In addition, scheduling and detailed planning also integrate any overlapping.

You have to enable the relevant configuration in the order type in order to verify the minimum send-ahead quantity when logging on OPs. Configure the processing code accordingly to plan overlapping operations based on the minimum send-ahead quantity (or the lead time). You can enable this function in the processing code while customizing the system.

When checking the minimum send-ahead quantity, the system only takes into account the recorded yield that has been entered up until now (primary quantity unit).

No quantity conversion takes place. For this reason, make sure that adjacent operations have the same primary quantity unit.

Example:

Operation 0100	Target quantity 1000	Send-ahead quantity 50
Operation 0200	Target quantity 1000	

If you enabled the validation check for the send-ahead quantity, you can only log on operation 0200, once operation 0100 has produced a yield quantity (in primary quantity unit) of at least 50.

The system does not check the operation status of the preceding operation. You cannot log on the current operation, in case the preceding operation has already been finished, but the send-ahead quantity has not yet been reached.

Target quantity (S and T) / unit / target scrap quantity (S and T)

The secondary and tertiary quantity are considered optional, variable units (for example within the reel-based MES solution - RF).

The indicated target quantity may include a target scrap quantity that might have been entered.

Target quantity (B) / unit / target scrap quantity (B)

The base quantity unit is an objective description of the material used in an order. The base quantity unit allows you to compare, for example, scrap from different operations. The base quantity unit is in effect the quantity unit shown in the order header. Generally, conversions (for example when target quantities are updated) are made using the base quantity unit.

The indicated target quantity may include a target scrap quantity that might have been entered.

If you use a quantity type, make sure to set the correct quantity unit.
The system only converts the quantities based on the conversion factors in the index tab "quantities" if the relevant values you want to recalculate are "empty" (not "0"). Quantity fields that contain values will not be recalculated.

Conversion factors

Use the conversion factors to convert the primary, secondary and tertiary quantities to the base quantity. Use these conversion factors, for example, when updating target quantities.

Use a numerator and denominator if you want to use decimal values (meaning a figure with decimal places) as the conversion factor.

Example

- Base quantity unit: Square meter M2
- Primary quantity unit: Piece PCE
- 1 piece = 2 square meters.

In this case
- define the numerator as the primary quantity 2 and
- define the denominator as the primary quantity 1

If no (valid) conversion factor exists, the system will attempt to convert the values using conversion formulas (this requires that formulas were defined during the customization process).

Overdelivery/ Underdelivery

The system checks all quantities you post for overdelivery. These quantities occur, when you report part quantities for an operation, when you interrupt or log off an operation. When you log off an operation, the system also runs a check for underdelivery.

For further information on overdelivery/underdelivery checking, see the document entitled MBL_PC_UnderOverDeliveryOverview.pdf.

Underdelivery (%)

Value shown as a percentage by which the quantity reported may deviate from the target quantity (primary quantity unit). The value is only assumed from the processing code if the value was not explicitly transmitted via the ERP interface.

Example:
Target quantity of the operation: 120 items
Underdelivery: 84%
The actual quantity must not fall below 101 items.

Overdelivery (%)

Value shown as a percentage by which the quantity reported may deviate from the target quantity (primary quantity unit). The value is only assumed from the processing code if the value was not explicitly transmitted via the ERP interface.

Example:
Target quantity of the operation: 120 items
Overdelivery: 168%
The actual quantity must not exceed 201 items.

Overdelivery reaction/ underdelivery reaction

If the limits specified in the fields *overdelivery* or *underdelivery* are exceeded, a warning or an error message may be issued in response. Possible values:

"empty" No reaction

W Warning

X Error.

If *error* is set as the reaction, you will not be able to override the validation check.

If *warning* is set as the reaction, you can override the validation check by entering a deviation reason.

Only Windows terminals allow you to enter a deviation reason. DOS terminals interpret the reaction "W" as an error.

Unit quantity

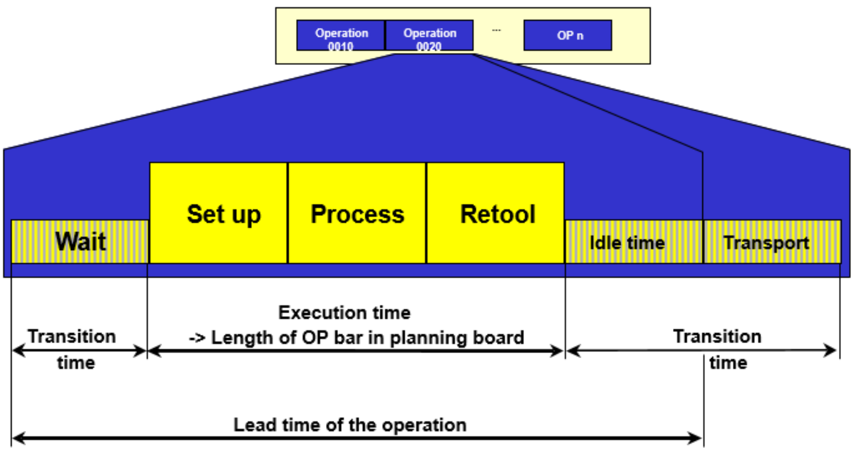
Quantity referring to operation specifications. You can customize the MES to use the ERP base quantity here. You can reference the ERP base quantity in formulas to calculate process times.

The unit of the unit quantity must be a primary quantity unit. The system does not perform an automatic conversion if the quantity units do not match.

As opposed to the base quantity in ERP, there is no other meaning or use in MES.

Durations / target times tabs

The illustration shown below provides an overview of the chronological structure of an operation in MES (in-house production).



Target setup time

The target setup time is the time required to prepare a workplace for the operation, for example the time needed to mount the necessary tools or to set the machine in compliance with the specifications ("setup time"). During this time, the workplace's capacity is shown as in use.

The ERP system transfers the target setup time or you can calculate the target setup time using a customized formula. The formula is based on default values. In this case, enter the formula in the field "setup time formula".

Additional setup time

The graphic detailed planning sets the additional setup time for the operation, if a setup change matrix is available and an additional setup time results from planning.
The additional setup time can also show a negative value.

Target processing time

The processing time is the time needed to process the material as part of an operation. During this time, the workplace's capacity is shown as in use. The processing time depends on the order quantity; it does include neither the setup time nor the retooling time.

The graphic detailed planning does not use the *processing time*. The graphic detailed planning calculates the processing time and/or remaining run time dynamically using the formula entered in the field "Formula RRT 1".

The ERP system transfers the target processing time or you can calculate the target processing time using a customized formula. The formula is based on default values. In this case, enter the formula in the field "processing time formula". Make sure to use the same basis to calculate the processing time and the remaining run time (field "Formula RRT1").

Planned retooling time

The planned retooling time (teardown time) is the time needed to reset the workplace back to its original state after the operation has been completed. This may require some tasks such as dismantling tools or performing some cleaning work. During this time, the workplace's capacity is shown as in use.

The ERP system transfers the planned retooling time or you can calculate the planned retooling time using a customized formula. The formula is based on default values. In this case, enter the formula in the field "Teardown time formula".

Planned delivery time

There is only one time component for external operations, i.e. the delivery time. The system synchronizes this time with the Gregorian calendar. The performance level has no relevance.

External processing

If this option is set, the operation is one that is performed externally. External operations are generally not planned, but only scheduled. In terms of lead time scheduling, for these kinds of operations the lead time only results from the delivery time.

If this option is not set, it will be considered an in-house operation. In this case, the following process times specify the capacity requirements (planning in HYDRA Shop Floor Scheduling):

- Planned setup time
- Planned processing time
- Planned retooling time

The following process times are used for scheduling (lead time scheduling) in-house operations:

- Target waiting time
- Target setup time
- Target processing time
- Target retooling time
- Target wait/idle time
- Target transport time.

Formula RRT1 / Formula RRT2

The value entered in the field *RRT 1* refers to a formula defined in the Management of formulas. The formula describes how to calculate the remaining run time (RRT) for an operation. The graphic detailed planning (HLS) uses this formula.

Unless otherwise specified or defined, enter the "RRT" value (remaining run time) in this field. In this case, calculate the remaining run time as follows (set by default):

(Target cycle / 1000) * (primary target quantity - the yield recorded up until now) / partitioning

You can enter another formula in the field *formula RRT 2.* You can use this formula to calculate any remaining run time that might deviate from *RRT 1.* The detail application *order progress* of the Order overview, for example, shows this formula.

Planned lead time

You can specify an overlapping of operations either using a send ahead quantity or lead time. The lead time describes the offset from the previous operation to its subsequent operation. A lead time can also be negative, if, for example, the subsequent operation begins with a setup before the previous operation.

Max. sync. time

If you enabled synchronization with the subsequent operation using the Processing code (customization), then planning makes sure that the maximum time span between this operation and the subsequent operation is the specified synchronization time.

The time is calculated in hours based on the shift calendar.

You can combine the synchronization function with an overlapping of operations.

Planned waiting time / waiting time formula / minimum waiting time

The waiting time is one available option to absorb interferences and delays for each operation. The waiting time describes the (calculated) length of time that needs to pass before an operation can commence (setup). The scheduling process also integrates the waiting time. You can enter the waiting time directly or you can calculate it using a formula.

You can reduce the waiting time during the scheduling process. For this purpose, you have to enter a reduction strategy for the order, which triggers a reduction in the wait time. You can reduce this time to the minimum waiting time (at most).

Target wait time

The target wait/idle time describes the length of time that needs to pass for processing-related reasons before a manufactured or processed material can undergo the next processing step. The scheduling process integrates the wait/idle time. You cannot reduce the wait/idle time.

Target transport time

The target transport time is the time necessary to transport material from one workplace to the next. The higher-level ERP system transfers the transport time or you can calculate the transport time in MES using a transport matrix.

Lead time scheduling takes into account the transport time. You can also reduce the transport time as part of lead time scheduling. You can define the transport matrix and reduction strategies when customizing the system.

Minimum transport time

If you use reduction strategies, you can reduce the transport time to this minimum amount of time during scheduling.

The following wage specifications are used to calculate an incentive wage.

Wage type

Wage type

Wage indicator

Piecework ID/premium (E/G/S/M/Z/...)

Target te

Premium default: te (per 1000 pieces).

"te" is the "time per unit" for each person. Use "te" to calculate the "order time", which is the specified processing time for each person used to calculate the incentive wage. By default, the MES shows this time in hours per 1000 pieces. The interface transfers this time in seconds per 1000 pieces.

If no incentive wage is used in MES, you can enter "0" here.

Target tr

Target tr is the person’s default setup time (in hours).

If no incentive wage is used in MES, you can enter "0" here.

Target teb

The premium default "teb" is the available machine time per unit. You can use this time to calculate the "occupancy time" for the workplace/machine so that the incentive wage can be calculated. By default, the MES shows this time in hours per 1000 pieces. The interface transfers this time in seconds per 1000 pieces.

If no incentive wage is used in MES, you can enter "0" here.

Target trb

Target "trb" is the default setup time (in hours) of the workplace/machine.

If no incentive wage is used in MES, you can enter "0" here.

Processing tab

Processing code

A processing code is a compilation of options that are used to control the operations. Each operation references this kind of processing code, and as a result its performance is defined in relation to the issues listed below.

You can define processing codes at the time the system is customized. Unless defined otherwise, enter the Processing code SYSTEM.

Recordable

If you set this option, you can generally post the operation, provided other criteria are also met (e.g. operation not locked or operation can be logged in due to the status of the previous operation).

Can be logged on at the same time/parallel logon possible

This option specifies whether an operation may be logged on several times, i.e. at the same time.

You should enable this option for overhead cost operations and for operations that are logged on to group workplaces. However, you should not set this option for operations that are subject to batch management.

The planning functions graphic detailed planning, order sequencing, graphic order sequencing do not support operations that can be logged on simultaneously. These planning functions assume that an operation is planned for exactly one capacity. If you log on one operation to several workplaces at the same time, contrary to capacity planning, this is then in opposition to planning. In order to conduct parallel planning of operations on different capacities, MES provides the operation splitting function.

Batch management requirement

Set this option if the operation is subject to batch management. You have to use the material management in order to process operations that are subject to management in batches.

Serial number requirement

You should only set this option after consultation with MPDV.

Layout

The code entered here references a label that was created in MES Label Designer that needs to be printed for the operation.

Target cycle

The target cycle is an operation-related specification used for machine clocking. The target cycle does not depend on the number of produced parts. In MES, the target cycle is calculated and processed as a duration per 1000 machine cycles.

If cycle time monitoring is active, this value is assumed as the default setting for finishing the operation. This value is the default value for the MDE machine monitoring function (cycle monitoring).

Partitioning

The partitioning (cavity) defines how many parts are produced during a machine cycle.

The partitioning is determined for each operation separately and is transferred via the ERP interface to MES. The partitioning is transmitted to the terminal at the time the operation is logged on and applies to the machine to which the operation is logged on.

Pulse factor

Automatic quantity collection at the terminal includes the pulse factor that is stored for an operation. Consequently, the pulse factor and the partitioning represent a conversion factor for the automatic collection of quantities: primary quantity = cycle * partitioning/pulse factor.

Split authorization

This option defines whether an operation may be split.

Max. number of splits

If an operation can be split, the system checks whether or not the split number entered by the user exceeds the value entered here. If this is the case, the split is rejected with an error.

M/O relation setup (machine/operator relation: setup)

Personnel requirements PEP (Personnel Scheduling) for setting up the operation.

Qualification (setup)

Unique qualification number from the qualification master data.

M/O relation production (machine/operator relation: production)

Number of employees required for production. By configuring the system during the customizing stage, you can define for each order type that only the number of persons specified here can log on. If several operations are logged on at the same time (in parallel), the maximum number of persons is equal to the total number of M/O relations for each separate operation.

In Personnel Scheduling (PEP), you can use this field to define the personnel requirements needed to produce the operation.

Alternatively to defining personnel requirements by way of the machine/operator ratio for the operation, you can also define these requirements in the production resources and tools. As opposed to the production resources and tools, you can only define one required qualification each for setup and production if the M/O relation is used.

The machine/operator relation (for setup and production) is only relevant for personnel scheduling if you have entered a qualification in the corresponding field.

Qualification (production)

Unique qualification number from the qualification master data.

Production method (variant)

Using production methods allows you to specify on which machine an operation can be planned, when the ERP system transfers order specifications. If you use the graphic detailed planning, you can apply the available production methods for detailed planning taking into account the specified times (target cycle, setup and retooling time) for each production method.

Here you can enter the key of the currently assigned production method.

Data identifier

Here, enter the data identifier if you use an Arburg Control System (ALS). This ID must be unique (key). If ALS is not used, leave this field empty.

CBM tab

This index tab is only relevant in connection with the reel-based solution using in the material management module.

General

Special indicators

Not used; this field must remain empty.

Number of reels

The planned total number of reels to be produced (parent roll and sub-rolls); no specific processing in MES.

Material properties

Input width

Reel input width in MM

Output width

Reel output width in MM

If multiple rolls are manufactured at the same time in one operation, this field indicates the sum total of the separate widths.

If branches are planned, the output width of the separate operations is set explicitly (no sum total is generated) for each operation ("parent" and "sub-roll" operations).

Seam width

Total seam width in mm

If several reels are produced at the same time in one operation, this field will contain the sum total of the separate seam widths.

If branches are planned, the seam width of the separate operations is set explicitly (no sum total is generated) for each operation ("parent" and "sub-roll" operations).

Surface per piece

Surface for a piece in MM2/PCE

Mass per unit area

Mass per unit area in G/MM2

Casing weight

This is where the casing weight for the sub-rolls is defined while cutting operations.
Unit: G

Cutting information

Cutting OP

Only relevant if the operation is a cutting operation

" " No roll cutting

"T" Roll cutting active (sub-roll numbering)

"M" Cutting active (parent rolls are being produced again)

Branch OP

Identifies parent and subordinate operations for a planned branch.

"M“ Mother OP of a planned branch

"K“ Child (subordinate) OP of a planned branch

Mother OP

If a branch is planned, these fields allocate the branched off material to the relevant mother operation.

A mother operation must reference itself.

Please note: Enter the MES order ID (= MES order number = combined order / operation number).

Daughter rolls/cut

Number of planned daughter reels per cut.

If the cutting plan is not defined, 0 is entered here.

Daughter rolls/cut - total

For cutting operations (mother OP): number of planned daughter rolls per cut (encompassing all branched off material).

If the cutting plan is not defined, 0 is entered here.

User fields tab

User fields allow you to store further customer-specific information to the MES in addition to the fields that are available by default. The order information shows operation-related user fields. The order information dialog includes the user fields index tab for the operations. This tab shows the user field key, the defined user fields including name and unit of measure. The user fields tab includes eight sub-index tabs, which each have eight additional user fields. The so-called user field key determines which user fields are involved and which meaning they have.

Object type

The object type for operation-related user fields is AGNR (cannot be modified).

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next. User field keys are defined in coordination with the customer during the customizing process.

User fields

The following user fields are available for the operation after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6

Field data type	Number of fields
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

Default values tab

You can define up to ten default values for each operation. You can use the default values, among other things, to calculate certain processing times using specified calculation rules. The default value key specifies the meaning of each separate default value .

Please note

We recommend **not** to change the default value key at the operation directly, because this might distort the meaning of the separate default values.

Default value keys are configured in coordination with the customer during the customizing process.

Administration tab

Created by/Created on

User who entered the operation and the time that the operation was entered. You cannot change these fields.

Modified by/Modified on

User who most recently modified the operation, and the time that this modification was made. You cannot change these fields.

Transferred by/Transfer time

Here, you can enter the source from where the operation was transferred. If the PPS system transfers the operation (PPS=J), the system automatically sets the transfer time and date to the time and date when the order was stored in MES. You cannot change these fields.

Modified HYDRA

Specifies that the operation was changed in MES. This identifier is automatically set at "J", if the OP was changed in MES. You cannot change the field.

Modified PPS

Specifies that the production order was changed in the ERP system. This identifier is automatically set at "J", if the production order was changed in the ERP system (PPS=J). You cannot change the field.

Deletion flag

This option is only displayed in the order information. You cannot change the option.

Responsibility area

If an area of responsibility is entered here, the user must have been authorized to view and edit operations and/or work plan operations.

The fields listed below are only displayed in the Order information. You cannot change these fields.

Locked / locked by / locked on

You cannot log in locked operations. The terminal does not show locked operations in the sequencing list, irrespective of how the status is configured.

Additionally, the user is shown who was the last to lock the operation and also the time and date when the operation was locked. These values remain even after the operation is unlocked. They are updated each time the operation is locked again.

Unlocked by/unlocked on

Shows the user who was the last to unlock the operation and the time and date when the operation was unlocked. These values remain even if the operation is locked again any time in the future. They are not updated until the operation has again been unlocked.

Locked for editing / locked for editing by / locked for editing on

Reserved; currently not used.

Reactivated by / Reactivated on

If an operation that has already ended is reactivated, the user is displayed here, who was the last to perform the reactivation and the time and date on which the reactivation took place.